



AI for Research: Methods and Practice

AI 

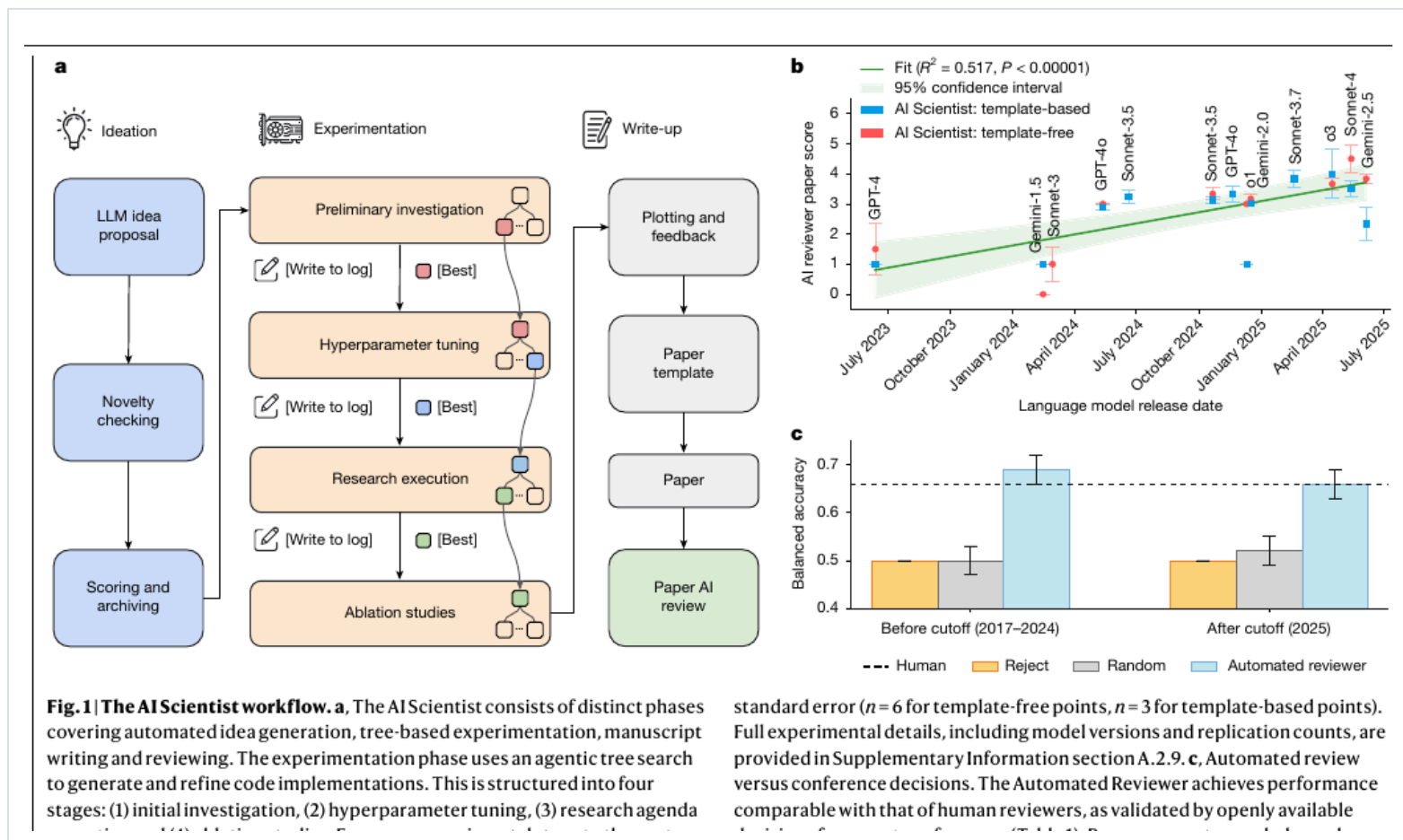
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Lu et al. · Nature 651 · 2026 · Fig. 1

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tree search

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Towards end-to-end automation of AI research

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The automation of science is a long-standing ambition in artificial intelligence (AI) research^{1,2}. Although the community has made substantial progress in automating individual components of the scientific process, a system that autonomously navigates the entire research life cycle—from conception to publication—has remained out of reach. Here we present a pipeline for automating the entire scientific process end to end. We present The AIScientist, which creates research ideas, writes code, runs experiments, plots and analyses data, writes the entire scientific manuscript, and performs its own peer review. Its ideas, execution and presentation are of sufficient quality that the manuscript generated by this AI system passed the first round of peer review for a workshop of a top-tier machine learning conference. The workshop had an acceptance rate of 70%. Our system leverages modern foundation models^{3–5} within a complex agentic system. We evaluate The AIScientist in two settings: a focused mode using human-provided code templates as an initial scaffold for conducting research on a specific topic and a template-free, open-ended mode that leverages agentic search for wider scientific exploration^{6,7}. Both settings produce diverse ideas and automatically test, report on and evaluate them. This achievement demonstrates the growing capacity of AI for making scientific contributions and signifies a potential paradigm shift in how research is conducted. As with any impactful new technology, there could be important risks, including taxing overwhelmed review systems and adding noise to the scientific literature. However, if developed responsibly, such autonomous systems could greatly accelerate scientific discovery.

AI has long been used to aid scientific discovery, an ambition with deep roots in the history of the field^{1,8–11}. Before the rise of large language models (LLMs), AI was limited to helping with specific, narrow tasks, such as discovering chemical structures², finding mathematical proofs⁵, discovering new materials^{12–14} and predicting the three-dimensional shape of proteins^{15,16}. Other systems focused on analysing pre-collected datasets to find new insights^{10,17,18}. However, with the recent advent of powerful and general foundation models, the role of AI has expanded to include assisting with a wider array of research activities. For example, LLMs now help with generating new hypotheses^{19–23}, writing literature reviews^{24,25} and coding experiments^{26–28}. Despite these advances in automating individual components, a system that autonomously navigates the entire research life cycle—from conception to publication—has remained out of reach until now.

This paper introduces The AIScientist, a pipeline that achieves the vision of full end-to-end automation of the scientific process. The AIScientist uses existing foundation models to perform ideation, literature search, experiment planning and implementation, result analysis,

A central challenge in developing such a system is automatically evaluating the quality of its scientific output at scale. To address this, we created an automated reviewer and first evaluated its performance against real, human-generated papers. The Automated Reviewer can accurately predict conference acceptance decisions, performing on par with human reviewers (Supplementary Information section A.3). We then used The Automated Reviewer to compare various configurations of The AIScientist by assessing how performance changes with the scale of the test-time compute and the quality of the underlying foundation model. We find that The AIScientist performs better with more compute resources (Fig. 3c). Furthermore, The Automated Reviewer shows that improvements to the base models significantly improve the quality of the generated papers, a finding that strongly implies that future versions of our system will be substantially more capable, as models continue to improve (Fig. 1b).

To assess The AIScientist in the same setting in which human-authored papers are evaluated, we conducted an experiment where we submitted generated papers to a workshop at the International Conference

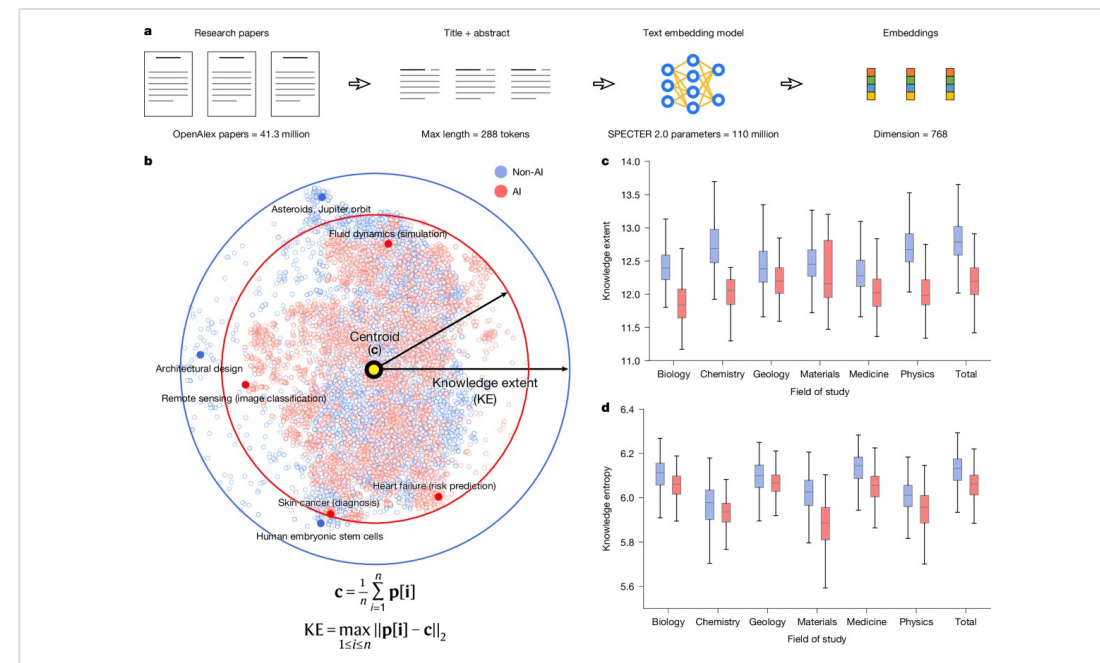
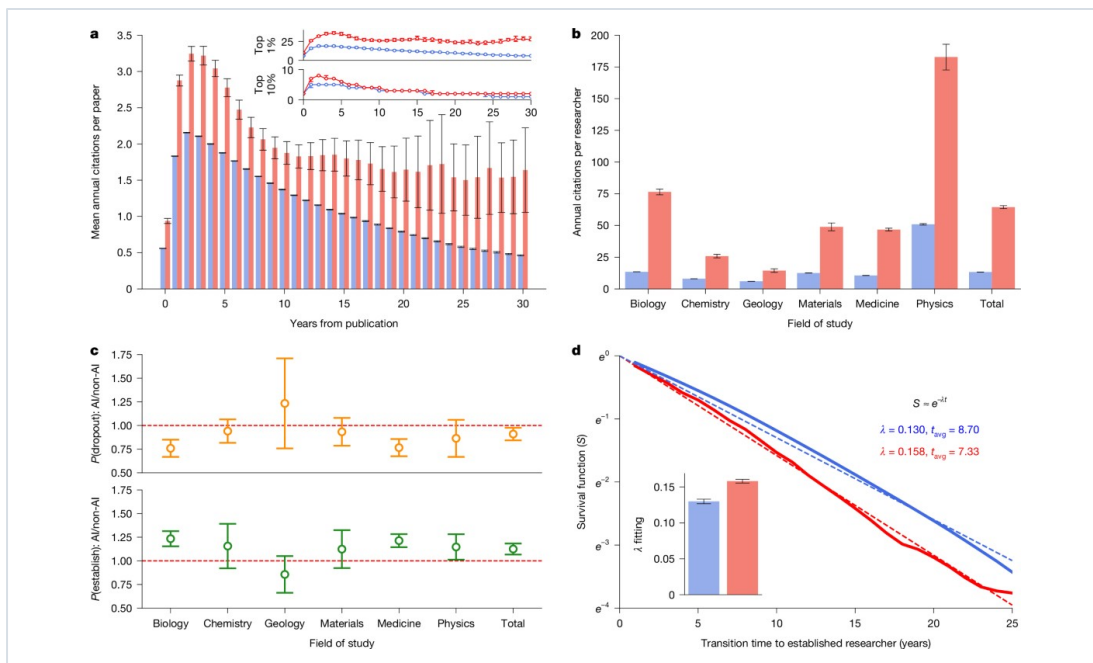
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workshop

70% ICBINB workshop

32% ICLR 2025

3 ICLR





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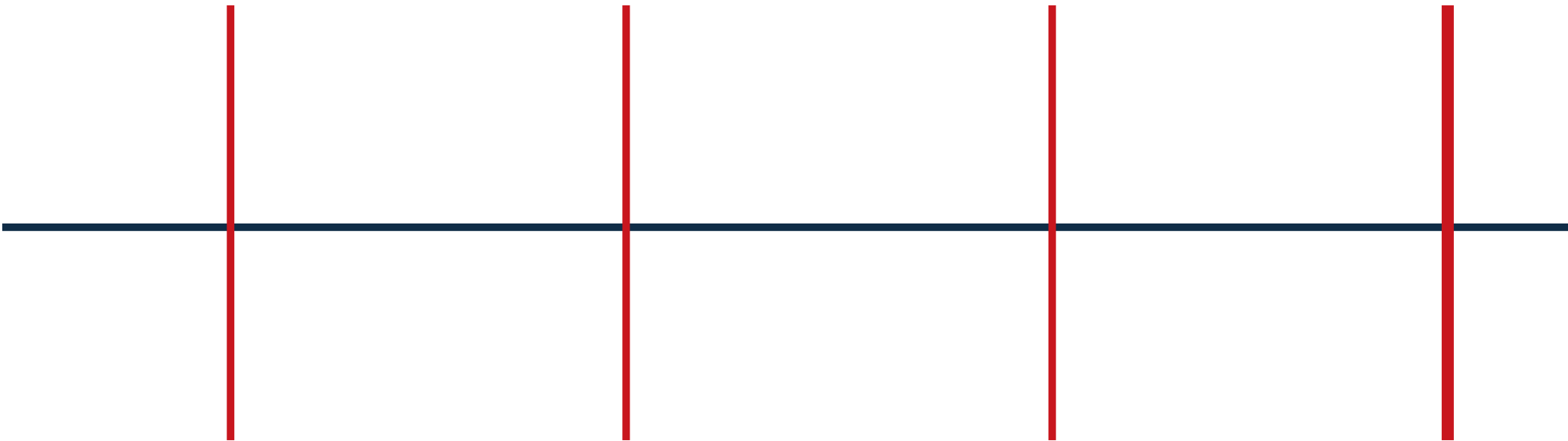


G1 · W6

G2 · W9

G3 · W13

G4 · W16



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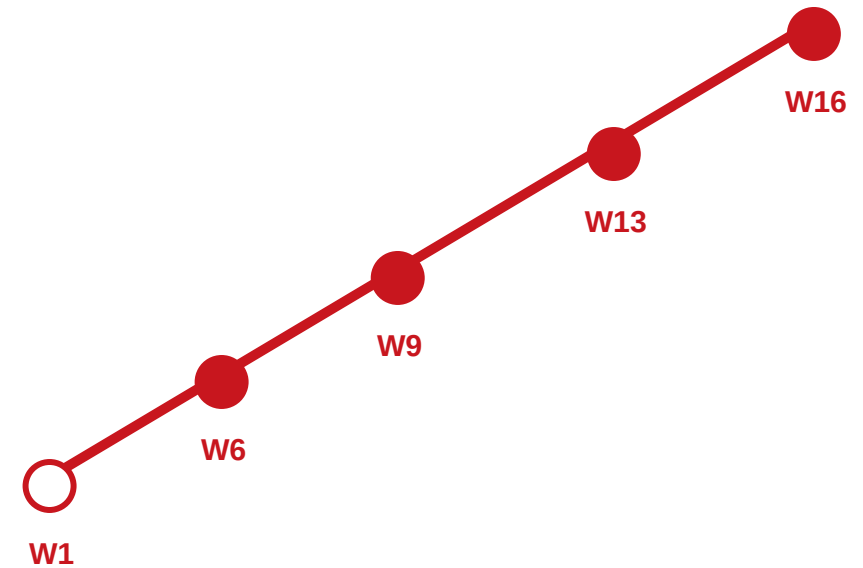


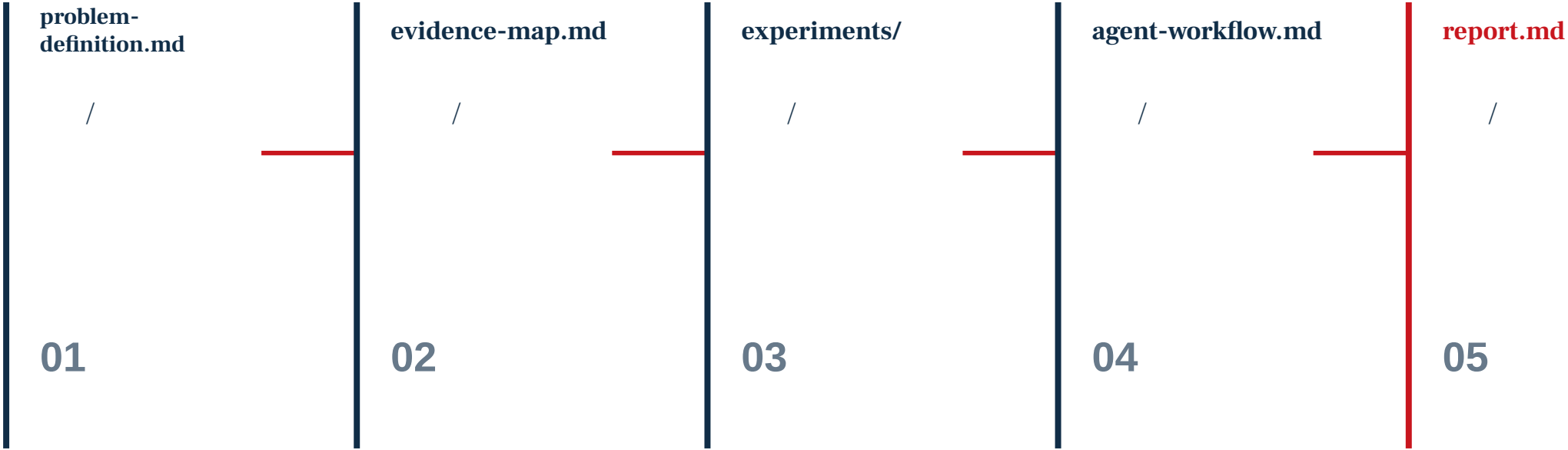


starter-template.md ·

```
project/
├── problem-definition.md
├── reading-cards.md
├── artifact-tracking.md
├── ai-usage-log.md
├── agent-permissions.md
├── ethics-and-compliance.md
└── notes/
```

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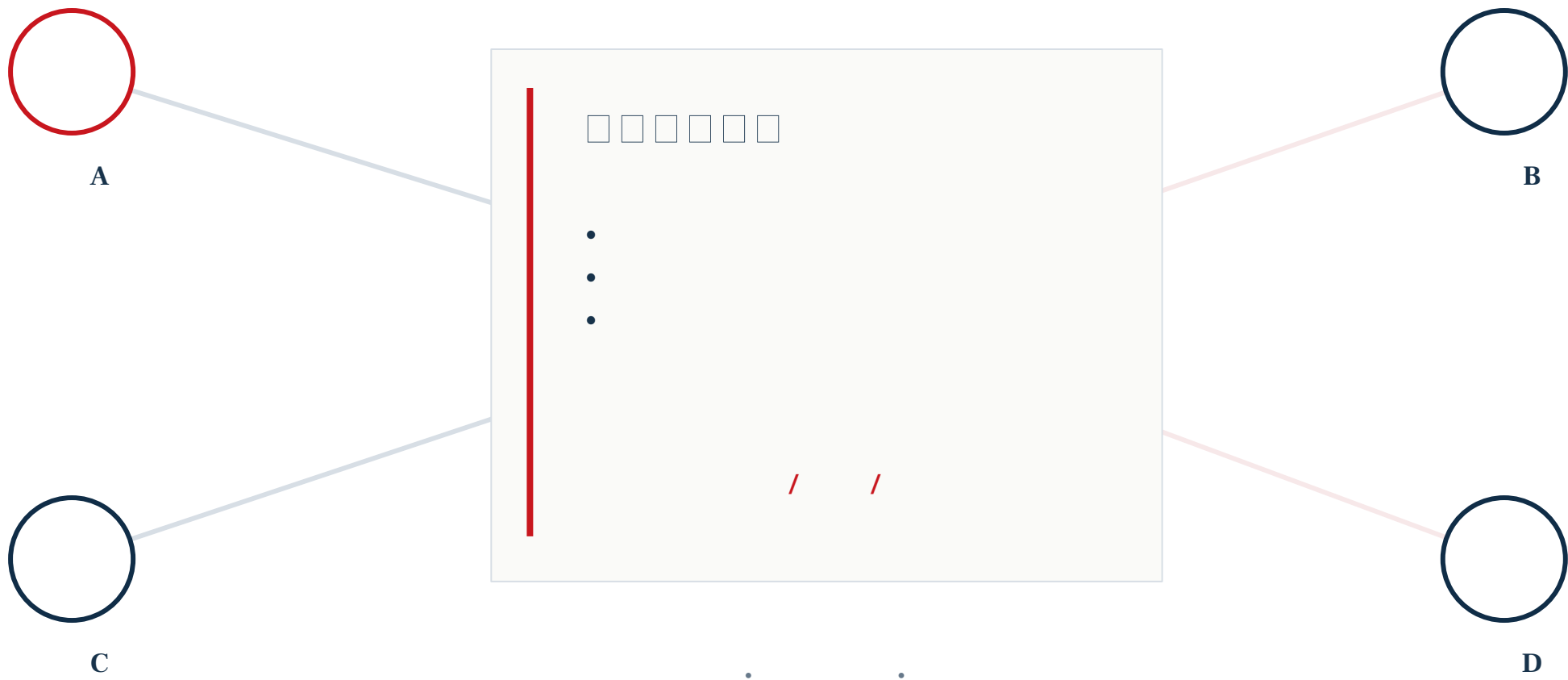






Agent Skills







problem-definition.md

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ai-usage-log.md

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Context _____

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